Woolworths Online Order and Delivery System – Enterprise Systems Analysis, Design, and Modelling

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Problem Statement

As online shopping continues to grow in popularity due to its convenience, Woolworths New Zealand has identified a need to implement an online system of its own (this is an assumption that Woolworths New Zealand does not have an online ordering and delivery system; the proposed idea is based on this assumption). Currently, Woolworths New Zealand does not have a dedicated online order and delivery system, which is a significant issue for multiple departments to perform effectively.

Customer service challenges
From the customer's perspective on the manual purchasing of groceries, there is no efficient way in which customers are able to compare and decide different product choices based on their nutritional value, price, and other product attributes. Customers would also not be able to access Woolworths online, as there is no system that provides the ability for customers to purchase goods. The customer would not have the convenience or mobility issues to be able to travel to a supermarket.

From the standpoint of a customer service representative, the absence of an online platform implies that customer service would have to be conducted in person, in-store for refunds, and deal with customer complaints, as well as have no organised method to manage digital customer complaints. This makes it difficult for Woolworths New Zealand to quickly address, track, and solve issues. This would erode consumer loyalty and confidence.

Operational challenges

The challenges that the store operation employees faced were that they were not able to visualize the total number of products offered in the store, thus needing to manually record and manage product stocks, which could increase the chances of human error. Employees would not be able to have statistical data on which product has the most demand, which leads to poor positioning of where products should be stocked.

Shift managers have no digital tools to delegate tasks or monitor the progress of staff, relying entirely on in-person communication. Store managers have no central system or efficient way for the store manager to track the performance of the store, which would help determine potential operational bottlenecks. Currently, the store manager relies on manual reporting, which is time-consuming and a possible task that could involve human error.

Supply chain and logistics impact

This issue would also affect the supply chain and logistics department, as procurement specialists are unable to make the most effective decisions in purchasing and sourcing products due to a lack of live visibility on real-time demand data. This provides inaccurate data to the procurement specialist, which would also affect the inventory controller, as they monitor and maintain optimal stock levels across store locations based on the procurement specialist's data. This would risk overstocking or understocking across various locations.

This also impacts delivery drivers, as ineffective supply allocation across store locations makes it difficult to plan and fulfil deliveries efficiently, raising overall operating expenses.

Proposed Solution

Woolworths New Zealand needs to develop an Online Order and Delivery System that allows customers to place an online order, in which customers are allowed to browse and search for products online. The customer should also be able to add products to their shopping cart and be allowed to check out. Customers would also be able to save a shopping list to reuse previous lists of orders to make it faster to shop for weekly used products. Customers will be able to create an account so that their details and saved shopping list are stored on the account for ease of access to previous orders or frequently purchased products. Customer service representatives should have access to a digital complaint and inquiry management system that allows them to log, track, and resolve customer issues efficiently, improving response times and maintaining customer loyalty.

In the store operation department, the store operation employee would be able to view digital information on the online system about what orders to pick and pack to prepare for delivery. The Shift Managers would use the system to delegate task on the store operation employees and be able to monitor their progress. The store manager would be able to have a view of performance metrics (tracking orders received, processed, and fulfilled), which would help identify bottlenecks or delays. Other task that is needed to be able to be performed include inventory oversight, reviewing customer feedback, and managing store information (display of online promotions, announcements, and opening hours).

For the supply chain and logistics department, the system should provide procurement specialists with access to real-time demand data, receiving messages about stock levels, especially if they are low, and track purchases made and the delivery of suppliers on the system. The procurement specialist should also be able to generate reports for planning future purchases. Inventory controllers should be able to monitor live stock levels across all store locations through the system, and monitor incoming stocks to ensure inventory is correctly recorded. Just like the procurement specialist, the inventory controller would also be able to generate reports on inventory to see which products need restocking the most and the least.

Delivery drivers should be assigned schedules automatically based on their availability and location, and should be able to update delivery statuses in real time so that customers are notified of their order progress. Drivers should also be able to flag issues on the system that need urgency.

Organisation Overview

Woolworths New Zealand (rebranded from Countdown) is a subsidiary of Woolworths Group Limited, an Australian publicly listed company. Woolworths New Zealand is a major supermarket chain whose headquarters are located in Mangere, Manukau, Auckland. Woolworth Group was founded by Percy Christmas on September 22, 1924, in Sydney, Australia. It is one of the 3 major supermarkets (Pak'n Save, Woolworths, New World) in New Zealand. Currently, there are 185 supermarkets[1] in New Zealand and 20,000 employees[1] working in New Zealand.

Woolworth's core values are providing convenience, value, range, and quality, and a motto of "The Fresh Food People"[2]. Thus, Woolworths needs dependable information systems to manage its operations across shops, distribution centers, and digital platforms because it depends greatly on customer satisfaction. One of these operations that would be focused on in this report is the Online and Delivery System, since it offers greater convenience and the ability to save time by ordering groceries online. These advantages in these systems align with the organisation's values, which are to provide the best experience for customers.

Departmental Structure and Responsibilities

In the ordering and delivery systems in Woolworths New Zealand, there are 4 critical departments in which forms the operation of the system. These systems are Store Operations, Supply Chain & Logistics, Customer Service, and IT.

- Store Operations

The store operations department for Woolworths New Zealand in 185 supermarkets[1] focuses on restocking shelves and maintaining the food freshness of merchandise. In the context of an online ordering and delivery system, they are responsible for managing orders online by picking and packing them for delivery.

- Supply Chain & Logistics

The Supply Chain & Logistics department for Woolworths New Zealand focuses on managing stock movement from the supplier and distribution centres to the store. In the context of an online ordering and delivery system, they are responsible for assigning deliveries to complete orders and managing their delivery schedule, as 22%[3] of orders online are fulfilled within 2 hours, and 57%[3] of orders online are fulfilled within the same day of ordering, as well as ensuring there is enough stock for deliveries.

- Customer Service

The Customer Service department for Woolworths New Zealand focuses mainly on making sure that all customers are satisfied with the service and merchandise by handling customer inquiries, complaints, and issues. In the context of an online ordering and delivery system, this department is responsible for handling customer complaints and issues, assisting customers online, and processing refunds for incorrect or out-of-stock merchandise on the online ordering and delivery system.

- IT

The IT department for Woolworths New Zealand focuses on maintaining all digital platforms and technology systems, ensuring that user data is secure, developing improvements to the website, and resolving technical issues. In terms of the context of the online ordering and delivery system, the IT department is responsible for maintaining the system, integrating inventory, supply chain, and payment systems.

- People and Culture (HR)

The People and Culture Department for Woolworths New Zealand focuses on making sure that all employees are well-trained, supported in their well-being, and manage payroll and recruitment. In the 2025 annual report, mental health programs such as 'Sonder' and leadership skills are completed by over 55,000[3] employees. This department also manages wage agreements and ensures there is an inclusive environment and culture.

- Finance

The finance department for Woolworths New Zealand is involved in managing the budget, managing the cash flow, planning, and analysing financial decisions, analysing financial risk, as well as producing financial reports. These responsibilities are important in overseeing the financial report of Woolworths New Zealand. Food's total sales reached $8,286 million[3] with normalised sales.

- Marketing

The marketing department for Woolworths New Zealand's main goal is to build brand awareness, build brand loyalty through rewards, and research the trends of the market, creating promotions and advertisements. This includes the marketing department managing the grocery shopping website, which contributed to having 2.1 million[3] active everyday reward members in Q4 of 2025.

Stakeholders and User Roles

Department	Stakeholder	Role

Store Operations	Store Operations Staff	Picks and prepares orders in the store
Store Operations	Store Manager	Oversees overall store operations and monitors system performance.
Store Operations	Shift Manager	Oversees store operations and staff delegation
Supply Chain and Logistics	Delivery Driver/Employee	Fulfils deliveries to customers
Supply Chain and Logistics	Procurement Specialist	Specialist Purchases stock for store locations
Supply Chain & Logistics	Inventory Controller	Manages stock distribution across store locations
Customer Service	Customer	Primary user of the online shopping platform
Customer Service	Customer Service Representative	Handles customer complaints and refunds

Information from Stakeholders

The methods used to gather information from stakeholders are mainly interviewing stakeholders with questions. As well as viewing similar systems in the same industry to find out what the existing common features are in the system.

Customer stakeholder questions (This response is from a friend acting as a customer stakeholder):

Question	Answer
What are some of the problems you have with the ordering process in existing systems from other organisations?	When ordering something, finding the product is hard. Sometimes the filter does not work. Adding items to the cart is slow. Item stock is not displayed properly (want real-time stock information).
How often do you currently shop for groceries, and what frustrates you most about the in-store experience?	The client said he went shopping once a week and that the thing that frustrates him the most is the checkout lines, which take too long.
What would make you choose online grocery shopping over visiting a store in person?	Prices should not be too expensive for online ordering, and people selecting items should be choosing quality products. The online experience should be nice and easy to use, such as not having too many unnecessary buttons (search bar, filter).
How would you prefer to be notified about your delivery status – SMS, email, or in-app notifications?	In app notifications, sms should be an option as another method of notifying.
Would access to exclusive online-only deals or Everyday Rewards points influence how often you shop online?	The stakeholder stated that if money is saved, filling out a form is not required. Then a deal or an everyday reward feature would be fine to have.

Shift Manager stakeholder questions (This response is from me as the stakeholder):

Question	Answer

How do you currently assign tasks to store operation staff, and what are the biggest challenges with this process?	Currently, the shift manager assigns tasks, having scheduled meetings daily on what each staff member's task is to complete, and the biggest challenge is having to organise a in person meet everytime since it is time-consuming.
How do you currently monitor the progress of staff during a shift, and how time-consuming is this?	To monitor the staff's progress, each staff will have a paper time sheet for each staff member, and when they are done with their shifts, they write down on that sheet. The sheets are collected at the end of every day to be evaluated by the shift manager.
What are the most common issues that arise during a shift that slow down operations?	Common issues are that it is hard to make any changes in plans, and you will need to meet in person to inform employees.
How are staff currently informed about which products need to be picked and packed for an order?	Staff are informed by getting a paper receipt on the physical board. When they pick and pack based on the first-come, first-served method.
How do you currently handle situations where a staff member falls behind on their tasks or is unavailable?	Additionally, help would be assigned to the task that is not up to schedule. Assess the situation for any improvements that could be made.

What features would be most useful to you in a system that allows you to delegate and monitor tasks digitally?	The feature where I could see all the store operation staff, tasks, and time spent towards the goal, as well as tracking the progress of each of them.
How would you prefer to view the progress of tasks – through a dashboard, notifications, or another method?	Prefer to view them through a dashboard with all the important features, metrics, and notifications when there is urgent information.

Procurement Specialist stakeholder questions (This response is from me as the stakeholder):

Question	Answer
What are the biggest challenges you face when sourcing and ordering products?	Identify which supplier is the most suitable for the budget and standards. As researching suppliers is done manually.
How do you currently track whether a supplier has fulfilled a purchase order?	This is done by inquiring with the supplier about their progress in fulfilling the order and tracking them on a spreadsheet to be checked manually on a daily basis.
How do you currently forecast how much stock needs to be ordered for each product?	Stock is forecast based on historical data, and purchasing decisions are made based on the sales record of the past rather than in real time.
How do you currently share procurement information with the inventory controller?	Once a supplier is selected, the information about the stock is written on a spreadsheet and shared with the inventory controller so that the inventory controller can manage the incoming

	stock.
What information would you need real-time access to in order to make better purchasing decisions?	Inventory, order tracking, real-time demand, Market & Pricing Data

System scope, boundaries, and assumptions

Scope

The system consists of two sides: the customer side, where it is the online ordering and delivery platform for customers, and the backend side, the internal management system that supports the operations of the staff. Three departments spans in the system, which include store operations, customer service, and supply chain & logistics. Each department would have varying levels of access through the internal management system.

Boundaries

In scope:

- Online ordering and delivery platform
- Internal staff management system across Store Operations, Supply Chain and Logistics, and Customer Service departments
- Real-time stock and inventory management
- Automated delivery scheduling and tracking
- Digital complaint and inquiry management

Out of scope:

- Payroll
- HR management
- financial accounting
- in-store sales

- supplier direct access to the system
- marketing and advertising

Assumptions

1. Assuming Woolworths has not implemented an order and delivery system and still relies on in-person grocery shopping.
2. Assuming standard industry benchmarks for staff roles, as I have no visibility into Woolworths' specific operating procedures.
3. All staff are assumed to have access to a device to use the internal management system.
4. Stock levels in the system are assumed to be accurate and up to date.
5. Payment is assumed not to be handled by the system and is processed through a 3rd party payment gateway.

Requirements

- [x] ~~Business goal~~
- [x] ~~User requirements~~
- [ ] Clearly defined functional requirements
- [ ] Clearly defined non-functional requirements
- [ ] contraints
- [ ] Traceability between stakeholders and requirements

Business Goals

Customer Service

- Improve customer convenience by providing an accessible online grocery shopping platform.
- Enhance customer satisfaction through effective delivery communication.
- Improve customer trust through an accessible online complaint and refund platform.

Store Operations

- Decrease the amount of in-person communication that shift managers and retail employees use.
- Reduce human error in store operations through digital task and order management.
- Increase stock management precision with real-time stock visibility.

Supply Chain & Logistics

- Use automated delivery scheduling to increase delivery efficiency.
- Incorporate real-time delivery status updates on deliveries.
- Improve forecasting accuracy by providing procurement and inventory staff with real-time data demands.

User Requirements

Customer should be able to:

- Browse and search for products online.
- Compare products based on nutritional value and price.
- Add products to a shopping cart and check out.
- Save a shopping list for reuse.
- Create an account to store their details and saved lists.
- View store promotions, opening/closing dates, and news.
- Track and receive notifications about their delivery status.
- Choose delivery options / pick up.

A customer service representative should be able to:

- Logging and managing customer complaints.
- Processing refunds for incorrect or out-of-stock items.
- Log and manage customer complaints digitally.

Store Operation Staff should be able to:

- View real-time stock and product inventory.
- Receive order information to pick and prepare orders for delivery.

Shift Manager should be able to:

- Delegate tasks to store operations staff through the system.
- Monitor the progress of staff tasks in real time.

Store Manager should be able to:

- Track overall performance.
- Track order fulfilment rates.
- Identify bottlenecks from analysing overall store operations.
- Oversee inventory levels in the store.

A delivery driver should be able to:

- Get automatically assigned delivery orders based on location and availability.
- Update the delivery schedule when assigned to a delivery.
- Flag urgent delivery issue.

A procurement specialist should be able to:

- Access real-time demand data to make informed purchasing decisions.
- Receive automated alerts when stock levels are low.
- Track purchases made and supplier deliveries through the system.
- Generate reports for planning future purchases.

An inventory controller should be able to:

- Monitor real-time stock levels across all store locations.
- Monitor incoming stock to ensure inventory is correctly recorded.
- Generate inventory reports to identify which products need restocking the most and least.

Functional Requirements

Customer

- The system must allow customers to browse and search for products by name, category, price, and nutritional value.
- The system must allow users to be able to view the products.
- The system must allow customers to add products to a shopping cart.
- The system must allow customers to remove products from a shopping cart.
- The system must allow customers to proceed to checkout from their shopping cart.
- The system must allow customers to create a personal account with their name, email address, password, and delivery address.
- The system must allow customers to view their order history through their personal account.
- The system must allow customers to save shopping lists from previous orders for reuse.
- The system must display the current delivery status of an order to customers, including order confirmed, order being prepared, out for delivery, and delivered.
- The system must automatically send delivery notifications to customers when their delivery status is updated.

Customer Service Representative

- The system must allow customer service representatives to log customer complaints and inquiries digitally.
- The system must allow customer service representatives to track the status of open complaints and inquiries.
- The system must allow customer service representatives to update and resolve logged complaints.
- The system must allow customer service representatives to process refunds for incorrect or out-of-stock items.
- The system must allow customer service representatives to send notifications to customers via in-app notifications or SMS regarding issues with their order.

Store Operation Staff

- The system must display the list of products and their quantities for each incoming order to store operations staff.
- The system must display the order priority status to store operations staff so that urgent orders are fulfilled first.
- The system must display the delivery deadline for each order to the store operations staff.
- The system must allow store operations staff to update the status of an order as it is being picked and packed.
- The system must display real-time stock levels and product inventory to store operations staff.
- The system must send a notification to store operations staff when a new order is assigned and requires picking and packing.

Shift Manager

- The system must allow shift managers to assign tasks to store operations staff.
- The system must allow shift managers to monitor the progress of assigned tasks in real time through a dashboard.
- The system must send a notification to the shift manager when an assigned task has not been updated within a defined time period.
- The system must allow shift managers to reassign tasks to different staff members when needed.

Non-Functional Requirements

Validate Requirements

Requirement UML Modelling

- [ ] Use Case Diagram covering full system scope
- [ ] Minimum of five fully dressed Use Case Descriptions
- [ ] Correct use of include, extend, and generalisation relationships

Initial Structural Design and Implementation

- [ ] Initial UML Class Diagram
- [ ] Mapping between UML classes and planned C# classes
- [ ] Identification of anticipated design weaknesses prior to OOP application
- [ ] OOP-based implementation in C# that strictly conforms to the UML models.
- [ ] In the Class diagram, provide reasons for each component by writing a short summary of each.

References

1. Woolworths. (2024). Online Supermarket: Online Grocery Shopping & Free Recipes at countdown.co.nz. Countdown NZ Ltd. https://www.woolworths.co.nz/info/about-us/our-company
2. O'Connell, J., & O'Connell, J. (2024, November 19). 1987 The Fresh Food People. Australian Food Timeline. https://australianfoodtimeline.com.au/fresh-food-people/
3. WOOLWORTHS GROUP LIMITED. (2025). Annual Report 2025. https://www.woolworthsgroup.com.au/content/dam/wwg/sustainability/reports/f25/Woolworths%20Group%20Annual%20Report%202025%20.pdf

Appendices

- GitHub (https://github.com/ShawnLeeyz/Woolworths-Systems-Analysis-Design-and-Modelling-Project-I.git)